

LIBERIA – AGRICULTURE CENSUS 2024 – METADATA REVIEW

1. Historical outline

The only previous Agriculture Census (AC) in the country was carried out in 1971. The present one, to which data and metadata correspond, is the second AC conducted in the country.

2. Legal basis and organization

Legal framework

The National Statistics and Geo-Information Act, approved on 22 July 2004, establishes that the Liberia Institute of Statistics and Geo-Information Services (LISGIS) conducts censuses in the country. Annex Two of the Act, titled “Scope, Content and the Time Schedule on Release of Statistical Data” establishes that an AC should be taken every five years.

Institutional framework and international collaboration

The AC 2024 was conducted by the LISGIS in collaboration with the Ministry of Agriculture (MA) and other key partners. A steering committee was formed, headed by the Minister of Agriculture and integrated by members from LISGIS, and different Ministries and institutions. A coordinating committee was created, including members from the Ministry of Finance and Development Planning (MFDP), Ministry of Internal Affairs (MIA), LISGIS, and different agricultural sector authorities and institutions. The census had a technical committee headed by the Assistant Director for Agriculture Statistics at LISGIS assisted by the Director for Statistics at the MA. The committee included members from LISGIS, MA, National Fisheries and Aquaculture Authority (NaFAA), Cooperative Development Agency (CDA) and MIA. The AC 2024 was funded by the Government of Liberia and the Harmonizing and Improving Statistics in West Africa Project (HISWA) from World Bank. FAO provided technical assistance, while a global partnership program led by the World Bank, FAO, and the International Fund for Agricultural Development (IFAD) through the 50x2030 Initiative provided technical guidance and methodological support.

Census staff

The AC 2024 included enumeration in the household sector, in the non-household sector and in communities. In the household sector, the AC involved 160 teams composed of one supervisor, four enumerators and one GIS specialist who also worked as an enumerator. In communities, data collection was conducted by 160 team supervisors. The data collection on non-household holdings was done by 30 enumerators, composed of 15 County Agriculture Coordinators (CACs) from the MA and 15 Directors of LISGIS County Statistics and Information Offices (CSIOs). The AC also had 11 regional coordinators, 22 national monitors, 30 data quality assurance officers (one in each county), and five HQ monitors who worked collaboratively to ensure full coverage and data quality. Data collection was supported by 18 field drivers and five monitoring drivers.

3. Reference date and period

Reference day: the day of the interview, for livestock number, holding identification and household members.

Reference periods:

- The agricultural season, approximately from 22 November 2022 to 22 December 2023, for parcels and plots, labour force, crops, fertilizers, pesticides; and
- The last 12 months, for machinery and equipment, livestock breeding.

4. Enumeration period

The LAC 2024 data collection was carried out from 22 January to 22 March 2024. The process faced challenges such as difficult terrain and poor internet connectivity, which caused delays in data synchronization. The COVID-19 pandemic indirectly affected AC activities. The frame for the AC's enumeration areas came from the National Population and Housing Census 2022 (NPHC 2022), that was delayed and finally carried out in 2022. This caused the AC to be delayed from 2022 to early 2024.

5. Scope of the census and definition of the statistical unit

The **census scope** covered agricultural production (crop and livestock), forestry, aquaculture and fisheries activities.

The **statistical unit** was the agricultural holding, defined as economic unit of agricultural production under single management comprising all livestock kept and all land used wholly or partly for agricultural production purposes, without regard to title, legal form or size. Two types of holdings were covered by the census: (i) holdings in the household sector ("agricultural households"), defined as a household with at least one member practicing own account agricultural activities (crop cultivation, livestock, aquaculture, fisheries, forestry production) which are one of the sources of the family's livelihood or income and that must have practiced those activities during the 2022/2023 farming year; and (ii) holdings falling in the above definition in the non-household sector.

Community-level data

Community-level data was collected in the AC 2024. The questionnaire was designed to collect localities-level data in all communities within the selected enumeration areas (EA).

6. Census coverage

Geographic coverage

The AC covered the entire country.

Cut-off threshold and other exclusions

No thresholds or other exclusions were used in the LAC 2024 in the household sector. Informal agricultural cooperatives and smaller farmer-based organizations (i.e., those having less than 25 acres) were excluded based on the definitions adopted.

7. Methodology

Methodological modality for conducting the census

The integrated census/survey modality was used in the AC 2024. As a beneficiary of the 50x2030 Initiative, this modality was a suitable option for the country which decided to conduct a sample-based census focused on structural and essential items. Other data (non-structural) that are needed will be incorporated in the follow-up modules of the integrated survey programme.

Relation to other censuses

The list of enumeration areas (EA) used in the AC 2024 was developed by the LISGIS in the National Population and Housing Census 2022 (NPHC 2022). The list included the number of agricultural households in each EA obtained from the agricultural activity section of the PHC questionnaire.

Frames

The sampling frame for the household sector consisted of the full list of EAs developed by LISGIS in the NPHC 2022. The sampling frame included the number of agricultural households, and it classified EAs into urban/rural areas. The sampling frame used for the non-household sector was developed using a mix of administrative sources and field data collection. The Cooperative Development Agency (CDA) provided a list of agricultural cooperatives, the National Bureau of Concessions provided a list of agricultural concessions, the Ministry of Internal Affairs provided a list of Communal Farms and the MA provided a list of Farmer Based Organizations (FBOs) and private farms. These lists were consolidated by LISGIS and a fifteen-day field listing verification and update exercise was conducted by the Census technical team. The final list resulting from the field verification and update exercise was used as a frame for the non-household sector. A decision was made to cover all holdings in the frame, since the number was relatively low. Also, field staff were instructed to include new holdings if they identify any.

Complete or/and sample enumeration methods

The AC 2024 used sample enumeration for holdings in the household sector and complete enumeration for holdings in the non-household sector.

Sample design

The sample design was a stratified cluster sampling with enumeration areas (EAs) as clusters and agricultural households as units of interest. A sample of 4 800 EAs was selected, and these EAs had a total of 269 652 agricultural households in the frame. The sample was allocated within strata (districts, urban/rural) proportional to the numbers of agricultural households computed in the frame. In total, about 78.8 percent of the sample was allocated to rural areas. The stratified sample of EAs was selected with probability proportional to the number of agricultural households. A complete listing of all households (both agricultural and non-agricultural) was carried out in the selected EAs, and detailed questions were addressed to all households that practiced agricultural activities during the farming season 2022/2023. The results of the AC 2024 are representative at the district level. A response rate of 92.8 percent was reported.

Data collection methods

The census data was collected using CAPI method with CSPro application. The EA maps were integrated into the CAPI application to enable field staff easily identify their assigned EAs.

Questionnaire(s) and items covered

Data were collected using two questionnaires, for agricultural households and for holdings in the non-household sector. The AC 2024 covered 21 out of the 23 essential items recommended in the WCA 2020.¹

The community survey questionnaire collected data regarding respondent characteristics, agricultural activities, livestock and poultry production, disasters and shocks, infrastructure and transportation, organizations, resources management, land prices and credit, key events, labour, producer prices, fisheries and inputs.

8. Use of technology

CAPI method was used for data collection. EAs maps were included in electronic devices. Power-Bi dashboard was used for real-time monitoring and tracking. Census results were disseminated online (LISGIS Website, NADA and Liberia Open Data Portal).

9. Data processing

Direct data capture was ensured by the CAPI method. Data was edited using CSPro programmes, with edit rules established by programmers and subject matter specialists at LISGIS and MA. In a few cases, manual editing techniques were applied to recode responses generated from “other (specify)” options. The SPSS software was used for this purpose. Data anonymization and archiving were done using Stata, R and Nesstar Publisher software.

10. Quality assurance

The data quality assurance began with the robust training of field staff. In addition to the training, several data quality assurance mechanisms were instituted: (i) the census technical team ensured that the necessary logical functions were integrated into the CAPI application to identify and prevent inconsistencies during data collection; (ii) the census technical team implemented a pretesting exercise of the census questionnaires as well as a pilot census, ensuring that the functionality and logical flow of the questionnaires and tablets were tested, and the necessary adjustments were made; (iii) the AC 2024 dashboard was setup to monitor the flow of data in real time enabling the census technical team to evaluate teams’ coverage of EAs and households; (iv) five HQ monitors were assigned to keep monitoring the dashboard and identifying any error in data submission from field teams, also, a robust field monitoring by National and Regional monitors as well as County Inspectors and Data Quality Assurance Officers ensured that data collectors were always in their assigned areas administering the census questionnaires; (v) enumerators easily received technical assistance to avoid potential errors

¹ The following essential items were not covered by the AC: (i) 0501 Type of livestock system; and (ii) 0503 Number of female breeding animals.

in data capture; and (vi) GPS coordinates were captured before and after the administration of the questionnaires allowing the census technical team to use GIS technology to ensure that teams visited all localities and households within their assigned EAs.

11. Data and metadata archiving

Metadata was published under DDI document version 1.1 on 10 March 2025 and is available at https://microdata.lisgislr.org/index.php/catalog/36/study-description#study_desc for agricultural households and https://microdata.fao.org/index.php/catalog/2710/study-description#study_desc1686767889862 for communities. Microdata was anonymized using statistical disclosure control methods. The process began with classifying variables into categories: variables to delete, quasi-identifiers, direct identifiers, and linked variables. All direct identifiers and unnecessary variables were removed. Quasi-identifiers and linked variables were analysed to formulate disclosure scenarios, and disclosure risk was measured using k-anonymity and probabilistic risk techniques. The primary anonymization methods applied included recoding, shuffling, and local suppression. Finally, information loss was assessed to ensure that anonymization methods did not compromise the dataset's utility. The anonymization process was conducted using the R software.

12. Data reconciliation

Data reconciliation was carried out for the non-household sector.

13. Dissemination of census results and microdata

Preliminary results for agricultural holdings in the household sector were published in 2024. The final report for agricultural households was published in February 2025 and is available at <https://microdata.lisgislr.org/index.php/catalog/36/download/138>. The final report presented data at the county level. Microdata for agricultural households was anonymized and is available for statistical and research purposes only under specific terms and condition at <https://microdata.lisgislr.org/index.php/catalog/36/get-microdata>. The final report for the non-household sector was published in August 2025 and is available at https://lisgis.gov.lr/censusreport/LAC_NonHH_FinalReport_09182025.pdf. Information not available in the final report and on the Open Data Portal may be obtained from LISGIS HQ.

14. Data sources and Availability

Institute of Statistics and Geo-Information Services (LISGIS). 2025. Liberia Agriculture Census 2024 - Community Operations. In: *LISGIS* [online]. Monrovia. [Cited 17 April 2025]. <https://microdata.lisgislr.org/index.php/catalog/35>

Institute of Statistics and Geo-Information Services (LISGIS). 2025. Liberia Agriculture Census 2024 - Household Listing. In: *LISGIS* [online]. Monrovia. [Cited 17 April 2025]. <https://microdata.lisgislr.org/index.php/catalog/36>

15. Contact

Liberia Institute of Statistics and Geo-Information Services (LISGIS)
Address: Executive Mansion Ground, P.O. Box, Monrovia, Liberia
Phone: +231 555-674890
WhatsApp: +231 775-860-349
Email: lisgis.admin@lisgis.gov.lr
Website: <https://lisgis.gov.lr/>